Module Summary: Reproducible Data Workflow with the NYC Airbnb Dataset

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Project: AI Programming Foundations — Reproducible Data Science Workflow

Overview

This project builds and documents a reproducible Python data workflow using the 2019 New York City Airbnb Open Data listings dataset, sourced from Kaggle (https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data). The workflow ingests the raw listing data with Pandas, applies two documented cleaning functions, computes exploratory summaries, and produces four labeled Matplotlib/Seaborn visualizations, all captured in a single executable Jupyter notebook (*data_workflow.ipynb*). The aim is to demonstrate a modular, transparent workflow that can serve as a foundation for later machine learning and deep learning projects on the same data.

Dataset Description

The dataset contains 48,895 rows and 16 columns, each row describing a single Airbnb listing in New York City as of 2019. Columns include a unique listing ID, host ID and host name, borough (*neighbourhood_group*) and neighbourhood, latitude/longitude, room type, nightly price, minimum-night requirement, number of reviews, date of the most recent review, reviews per month, the host's total listing count, and yearly availability. This project focused primarily on *price*, *room_type*, and *neighbourhood_group* as the key variables for understanding what drives listing cost, using the review-related fields as a secondary indicator of listing activity. Because the data is a public snapshot of self-reported host listings rather than verified transaction records, it reflects the NYC short-term rental market only as it stood in 2019.

Workflow Description

The notebook follows a linear, five-stage workflow. **Ingestion** loads the CSV with Pandas and confirms the shape, column types, and first rows. **Cleaning** applies two purpose-built functions: one resolves missing values, and one removes invalid or extreme outlier rows. **Exploratory analysis** runs a single function that returns descriptive statistics, average price and listing counts by borough, and a correlation matrix of the numeric features. **Visualization** produces four labeled plots covering the price distribution, average price by borough, price by room type, and the geographic footprint of listings. **Summary** closes the notebook with a markdown narrative describing the patterns found, along with the assumptions and limitations of the analysis.

Key Decisions and Assumptions

Cleaning choices. Missing *reviews_per_month* and *last_review* values were treated as structurally missing — they line up almost exactly with listings that have zero recorded reviews — so *reviews_per_month* was filled with 0 rather than an imputed average, and *last_review* was left as a missing date rather than a fabricated one. Rows with a listed price of $0 were dropped as clear data errors, and rows above the 99th percentile of price ($799) or minimum nights (45) were removed as extreme outliers that would otherwise distort summary statistics and compress the visible range of typical listings in the plots. Percentile cutoffs were used instead of fixed thresholds so that the cutoffs reflect the actual shape of this dataset. Keeping each cleaning step in its own function with a clear docstring follows the tidy-data principle that a dataset's values, and the operations performed on it, should be easy to inspect and reproduce independently (Wickham, 2014).

EDA focus. The exploratory analysis intentionally centered on borough and room type, since these are the two categorical variables most directly under a host's control and most likely to explain price variation in a short-term rental market. **What each plot was designed to show:** the price histogram (Figure 1) shows the overall shape and skew of pricing after cleaning; the borough bar chart (Figure 2) isolates the geographic price effect; the room-type box plot (Figure 3) isolates the effect of unit type and shows within-group spread; and the geographic scatter plot (Figure 4) sanity-checks the cleaned data against the real shape of New York City and shows where listing density is highest.

Results and Interpretation

After cleaning, 47,964 of the original 48,895 listings (about 98%) remained. Figure 1 shows a right-skewed price distribution: most listings fall under $150 per night, with a long but much thinner tail stretching toward the $799 cutoff. Figure 2 shows a clear borough effect — Manhattan listings average roughly $173/night, well above Brooklyn (~$116), while the Bronx, Queens, and Staten Island cluster together in the $84–$94 range. Figure 3 shows that entire homes/apartments are priced well above private or shared rooms and have the widest spread, consistent with that category ranging from modest apartments to large luxury units. Figure 4 confirms that listings are heavily concentrated in Manhattan and Brooklyn, with much sparser coverage in the outer boroughs, mirroring the listing counts behind Figure 2. The correlation matrix from the exploratory-analysis function showed only weak linear relationships between price and the review-activity fields, suggesting that how often a listing is reviewed says little about what it costs; borough and room type are the stronger, if categorical, drivers of price in this dataset.

Responsible Practice (Bias and Data Quality)

Two cleaning decisions carry the most risk of introducing bias. Removing the top 1% of listings by price and by minimum-night requirement excludes real luxury and likely long-term or corporate rental listings; left unexamined, this could make the NYC market look more price-homogeneous than it actually is and could understate the extent of price inequality across listings. Filling missing review fields with zero assumes every missing value reflects a genuinely unreviewed listing; if any of that missingness instead reflected a data-collection gap, the analysis would be silently treating those listings as less active than they may actually be. Both risks were mitigated the same way: every threshold and assumption is stated explicitly in the notebook, in the code's docstrings and in a companion "Cleaning Justification" note, so a downstream user can see exactly what was changed and re-run the analysis with different assumptions if needed. This kind of explicit documentation of preprocessing choices is consistent with recommended practice for reproducible, ethically sound data workflows (Danchev, 2022).

Reproducibility

The project is reproducible in two complementary ways. First, *requirements.txt* was generated with pip freeze from the exact virtual environment used to run the notebook, pinning every package to the version actually used; a reviewer can recreate that environment with pip install -r requirements.txt before running the notebook, the same environment-pinning approach used to make the companion learning resource on reproducible data science reproducible via MyBinder (Danchev, 2022). Second, the project is tracked in a GitHub repository with multiple commits documenting progress and at least one additional development branch beyond *main*, so the history of changes to the notebook, cleaning functions, and report is auditable rather than only reflected in a single final snapshot.

Sources and Citations

In-text citations above follow APA style. Full references:

Danchev, V. (2022). Reproducible data science with Python: An open learning resource. *Journal of Open Source Education, 5*(56), 156. https://doi.org/10.21105/jose.00156

Wickham, H. (2014). Tidy data. *Journal of Statistical Software, 59*(10), 1–23. https://doi.org/10.18637/jss.v059.i10

Kaggle. (n.d.). *New York City Airbnb open data* [Data set]. https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data